

A Modern Ecommerce System — Design

Bounded contexts, aggregates, commands, and context relationships (sequence & architecture views)

The system is decomposed into eight bounded contexts around ownership and change-rate boundaries. Checkout is orchestrated by a process manager so that stock reservation, payment authorization, and capture stay strictly ordered; everything downstream (fulfillment, notifications, search projections) is choreographed through domain events. This keeps the design consistent with an event-driven, DDD-first approach while remaining a single deployable unit for a small team.

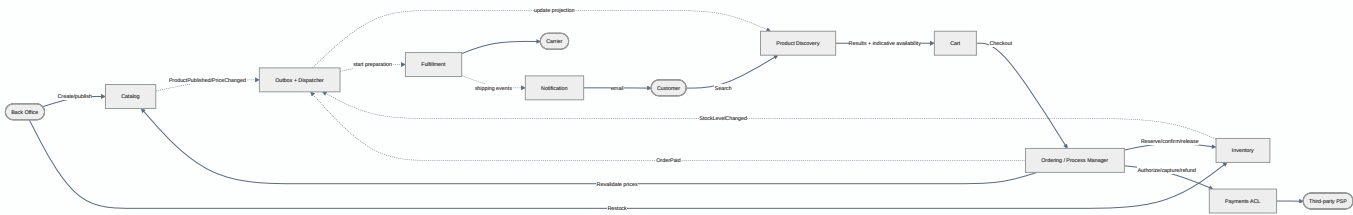
1 Bounded Contexts

Table 1 — Context ownership

Context	Owns (and explicitly does not own)
Catalog	Product data, media, list prices, publishing — <i>not</i> stock levels
Inventory	Stock per product, reservations, restocking, non-overselling invariant — <i>not</i> prices, orders, payments
Cart	Cart lines and price snapshots — <i>not</i> availability guarantees
Ordering / Checkout	Orders, state machine, checkout orchestration, compensations — <i>not</i> payment details or physical fulfillment
Payments	PSP abstraction: authorize / capture / void / refund, reconciliation — <i>not</i> stock
Fulfillment	Preparation and shipping of paid orders, tracking — <i>not</i> payments or pricing
Product Discovery	Search index (name + description), filters, indicative availability — read-only projection
Notification	Transactional emails / status updates — <i>not</i> physical fulfillment

Figure 1 — Context relationships and event flow

Solid = synchronous call. Dashed = asynchronous domain event via outbox.



2 Aggregates, Entities, and Commands

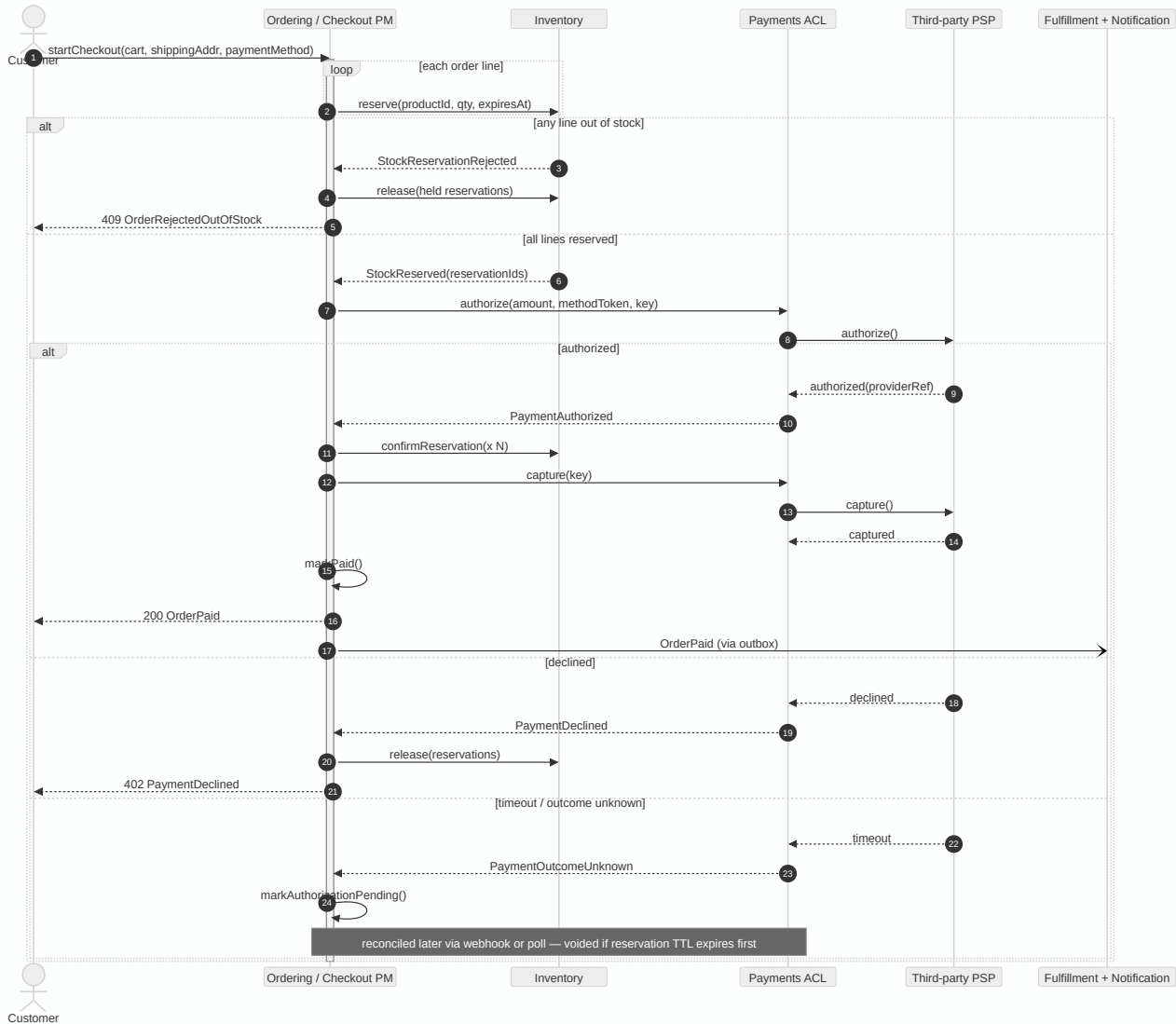
Table 2 — Aggregate roots, key commands and invariants

Aggregate	Commands → emitted events	Key invariant
Product Catalog	create · attachMedia · updateDetails · changePrice · publish · discontinue → Product{Created,DetailsUpdated,Published,Discontinued}, PriceChanged	A published product has a name, description, price, and ≥1 image
StockItem Inventory	receiveStock · reserve · confirmReservation · releaseReservation · reverseConfirmation · adjustStock → Stock{Received,Reserved,Confirmed,Released,ConfirmationReversed} + StockLevelChanged	available = onHand - reserved ≥ 0
Cart Cart	addItem · changeQuantity · removeItem · mergeIntoCustomerCart → CartUpdated, CartMerged	Positive quantities, one line/product, not expired or already in checkout
Order Ordering	startCheckout · confirmAllStockReserved · markPaymentAuthorized · confirmStock · markPaid · rejectOutOfStock · cancel → CheckoutStarted ... OrderPaid / OrderRejectedOutOfStock / OrderCancelled	No payment without an active reservation per line; Paid requires confirmed stock + captured payment; totals frozen once authorization starts
Payment Payments	authorize · capture · voidAuthorization · refund → PaymentAuthorized/Declined/OutcomeUnknown, Captured/CaptureFailed, AuthorizationVoided, Refunded	Retries never duplicate a charge (idempotency key); active captures ≤ order total
Shipment Fulfillment	prepare · markShipped → ShipmentPrepared, OrderShipped	Created only for a paid order; cannot ship without tracking info

3 Checkout Sequence — Happy Path, Out-of-Stock, Payment Timeout

Figure 2 — Orchestrated checkout (one flow, three outcomes)

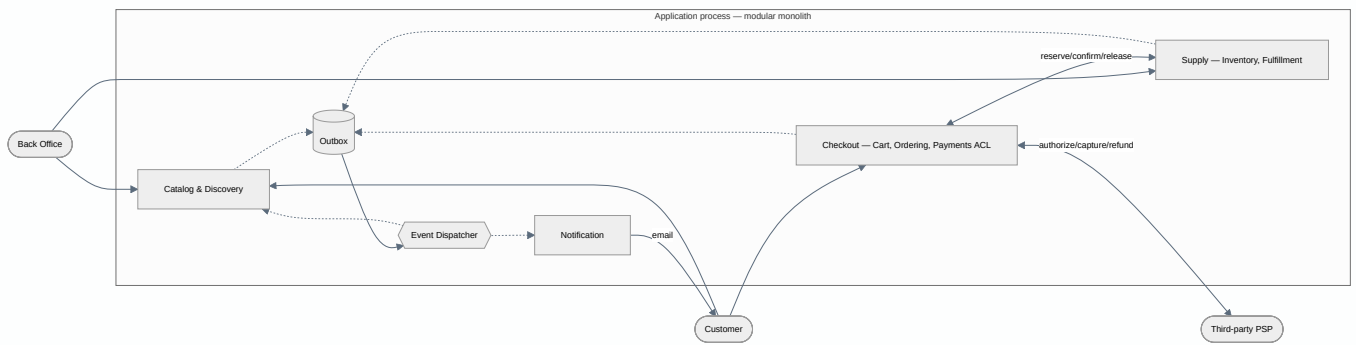
Reservation always precedes authorization; authorization always precedes capture. Out-of-stock and payment-timeout are modeled as alternative branches of the same interaction, not separate flows.



4 Physical Architecture

Figure 3 — Modular monolith, one deployable process

One module per team; module-owned schemas; a durable outbox/dispatcher instead of an in-memory bus, so the same code can later publish to an external broker.



Team ownership: **Catalog & Discovery** → Catalog, Product Discovery · **Checkout** → Cart, Ordering, Payments · **Supply** → Inventory, Fulfillment · **Customer Comms** → Notification. Modules share no aggregates; the release train is shared, internals are not.

5 Edge Cases and Requirement Coverage

- **Partial reservation failure** — already-held reservations are released before responding; no partially-valid multi-product hold. Reservation TTL is the safety net if compensation itself fails to run.
- **Ambiguous payment outcome** — a timeout is not a decline; webhook + polling reconcile via `ProviderReference` / `IdempotencyKey`; a late authorization is voided, never captured, if the reservation already expired.
- **Capture/confirmation failure** — authorization is voided and confirmed reservations are compensated via `reverseConfirmation`, both idempotent and audit-logged.
- **Price drift** — prices are revalidated between add-to-cart and checkout; totals become immutable once authorization starts.

Table 3 — Requirement coverage

Requirement	Mechanism
Search by name/description; add to cart if available	Search projection (Fig. 1); best-effort <code>inStock</code> check, authoritative reservation at checkout (Fig. 2)
Back office: product + image/description/qty; restock	<code>Product</code> and <code>StockItem.receiveStock()</code> (Table 2)
Checkout with address + payment method, guest allowed	<code>Order.Buyer</code> : required email, optional <code>customerId</code>
Payments via third party; stock kept independent of payments	<code>Payments</code> ACL never reads stock; <code>Inventory</code> never calls the PSP
Out of stock blocks payment and is shown in the UI	Payment is only ever called after all reservations succeed (Fig. 2, branch 1)
Order preparation, shipping, and notifications	<code>Fulfillment</code> + <code>Notification</code> , event-driven off <code>OrderPaid</code> (Fig. 1)

Scalability comes from read projections and product-level write isolation; resilience from outbox/inbox, idempotency, reconciliation, TTLs, and compensations. Trade-off: abandoned checkouts hold stock temporarily and availability can be briefly stale — accepted to avoid distributed transactions. Modules should only be extracted into services when load, compliance, or team autonomy actually require it.